

AI-Powered Multimodal Disaster Intelligence and Victim Detection System for CSSR

Operations:

Problem Statement:

Description: Rescue teams lack unified, real-time disaster intelligence

During disasters such as earthquakes, building collapses, floods, landslides, industrial accidents, and fires, rescue teams face critical challenges in locating victims quickly. Traditional operations rely on manual searching, CCTV footage, drones, or isolated sensor systems, which often fail to deliver timely and accurate results.

Proposed Solution: The proposed system is an AI-based disaster management platform that integrates image analysis, video analysis, audio signal processing, and text intelligence to identify trapped victims, assess disaster severity, and provide rescue insights in real time.

- Image and video analysis to detect victims, fire, smoke, and structural damage from drone and CCTV footage
- Audio signal processing to identify human screams, distress calls, and emergency sounds in noisy environments
- Text intelligence using natural language processing to extract victim location, count, and urgency from SOS messages and social media
- A multimodal fusion engine that combines outputs from all four modalities to generate high-confidence rescue alerts
- A real-time rescue dashboard providing danger zone heatmaps, priority rankings, and decision support for rescue teams

Dataset:

The dataset consists of two categories: **Victim** and **No-Victim**.

- The **Victim** folder contains disaster-scene images with visible victims along with corresponding JSON annotation files that define victim locations and boundaries.
- The **No-Victim** folder contains disaster-scene images without any victims and does not require annotations.

Dataset Link: [Disaster detection Frames - Google Drive](#)